A DOMAIN WITH FACTORIZATIONS INTO IRREDUCIBLES IS A UFD EXACTLY WHEN EVERY IRREDUCIBLE IS PRIME

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Unique factorization read on products of irreducibles. Used in the first half of the counted proof (lines 29651-29660). Lets a single irreducible appearing in a product be recognised as an associate of one of the factors.
- Induction on the total number of irreducible factors. Used in the second half (lines 29666-29671). Reduces uniqueness of two factorizations to the case of one fewer factor on each side.

Lemma 0.1. *Let R be a domain. Assume every nonzero, nonunit factors into irreducibles. Then R is a UFD if and only if every irreducible element is prime.*

Proof. Assume R is a UFD and let $x \in R$ be an irreducible element. Say $ab \in (x)$, i.e., $ab = cx$. Choose factorizations $a = a_1 \ldots a_n$, $b = b_1 \ldots b_m$, and $c = c_1 \ldots c_r$. By uniqueness of the factorization

$$a_1 \ldots a_n b_1 \ldots b_m = c_1 \ldots c_r x$$

we find that x is an associate of one of the elements $a_1, \ldots, b_m$. In other words, either $a \in (x)$ or $b \in (x)$ and we conclude that x is prime.

Assume every irreducible element is prime. We have to prove that factorization into irreducibles is unique up to permutation and taking associates. Say $a_1 \ldots a_m = b_1 \ldots b_n$ with a_i and b_j irreducible. Since a_1 is prime, we see that $b_j \in (a_1)$ for some

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j. After renumbering we may assume $b_1 \in (a_1)$. Then $b_1 = a_1 u$ and since b_1 is irreducible we see that u is a unit. Hence a_1 and b_1 are associates and $a_2 \ldots a_n = u b_2 \ldots b_m$. By induction on $n + m$ we see that $n = m$ and a_i associate to $b_{\sigma(i)}$ for $i = 2, \ldots, n$ as desired. $\square$